

Amr Shafik

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Professional Summary

Dedicated researcher with a PhD in Civil and Environmental Engineering from Virginia Tech, specializing in transportation systems engineering. Expertise in eco-driving for connected vehicles, traffic signal optimization, and traffic state estimation. Skilled in simulation analysis, predictive modeling, and optimization, using Python, R, C++, and Java. Industry and academic experience include large-scale mobility projects with the World Bank and EBRD. I am currently working as a research Engineer at the Center of Applied Transportation Sciences where I have the opportunity to apply and expand my expertise in real world problems, and to graduate students and advancing research in transportation modeling and software development.

Work Experience

Research Engineer

Arizona Transportation Institute (AZTI), Tucson, AZ

March 2025 - Present

**University of Arizona, Center of Applied Transportation Sciences (CATS),
Tucson, AZ**

AZTI Duties:

- Coordinating Proposal Evaluation Committee meetings for the Arizona Department of Transportation.
- Working with AZTI staff to help manage the research program for AZTI and providing key inputs regarding the program's growth.
- Review the scope of work, tasks, and deliverables across all AZTI research projects
- Coordinating the Annual Transportation Research Summit (2025) in Flagstaff, AZ.
- Project Manager for the Inexperienced Driver Mitigation (ABOR-funded Project).
- Coordinating between stakeholders (such as ADOT), AZTI faculty, AZTI staff, and leadership.
- Stakeholder coordination and engaging in project meetings with ADOT representatives, TAC members, and other partners to discuss progress, results, and implementation paths.
- Promotes AZTI and disseminates its initiatives at national and international conferences.
- Help train lower-level staff

CATS Duties:

- Solving real-world transportation challenges through applied research.
- Conduct advanced research in transportation modeling, traffic control, and intelligent transportation systems.
- Supervise and mentor graduate students while collaborating on large-scale mobility projects.
- Project Manager for the following projects:
 - Traffic Engineering Technical Support for the City of Yuma, Arizona
 - Inexperienced Driver Mitigation (ABOR-funded Project)
 - Helping to increase funding for the Center of Applied Transportation Sciences.
- Participated in the research Team for the following projects:
 - NCHRP 07-34: Artificial Intelligence for Transportation Systems Management and Operations Applications.
- Participates in the writing and research for publications and presentations:
 - Develop and write grant proposals and prepare reports of research results for publications and presentations and contribute to securing external research funding.
 - Plans, organizes, and conducts research projects, including methodologies, databases, and protocols, using innovative approaches to solve complex problems.

Traffic Engineer

District Department of Transportation (DDOT) - Summer Internship,
Washington, DC

June 2023 – August 2023

- Reviewing Traffic Impact Technical Reports and Traffic Analysis Models:
 - Assessed traffic impact reports and traffic management plans for I-395 in Washington, DC, aimed at mitigating congestion due to bridge rehabilitation projects.
 - Reviewed Synchro models for multiple consecutive signalized corridors in Washington, DC.
- Conducting Site Visits to Monitor Project Progress:
 - Performed site visits and participated in on-site follow-up meetings with consultants and contractors to discuss current progress and address challenges at the Florida Ave./New York Ave. intersection in Washington, DC.

Freelance Transportation Planning and Traffic Engineer

Jan 2017 – Aug 2021

Highlight Projects:

- Replicated and updated Egypt's national transportation demand model, originally developed in CUBE software by the Japan International Cooperation Agency (JICA) in 2010, to incorporate current demand patterns, for both passengers and freight movements.
- Calibration of demand models using traffic survey data collected at several projects of railway and roadway feasibility studies.
- Developing macroscopic transportation models for various railway and road strategic projects using "PTV Visum Software".
- Analyzing traffic impact assessment of new developments for more than 10 projects in Egypt and the gulf region, including schools, shopping malls, residential neighborhoods, and educational and healthcare facilities.

Design Engineer

Hesham Hamed Consulting Office

Aug 2016 – Jan 2017

Giza, Egypt

Main Duties:

- Steel Structure design – sub consulting for Tiger Steel Engineering in the UAE.
- Develop steel connection design documents.
- Coordinating and communicating with clients in the UAE about revisions in design and requirements.

Technical Skills

- Software:
 - AutoCAD 2D and Road Design using AutoCAD Civil 3D.
 - Strong proficiency in Transportation modeling, Traffic Analysis and Simulation: PTV Vissim, PTV Visum, Synchro, HCS, SIDRA.
 - GIS: ArcMap, ArcGIS Pro, QGIS.
 - Using all Microsoft Office Software and LaTeX
- Programming:
 - Basic coding knowledge using: C++, Java, VBA, MySQL.
 - Proficient in Python, R.
 - Machine Learning framework: TensorFlow, Keras.
- Modeling and Optimization:
 - Transportation Network Modeling.
 - Transportation Demand Estimation and Modeling.
 - Traffic Simulation and Analysis.
 - Traffic Survey Analysis.

- System Optimization:
 - Building Optimization Models using CPLEX, Gurobi.
 - Optimization of Traffic Control Systems (Such as Traffic Signals and Public Transit).
- Languages Spoken:
 - Arabic (Native).
 - English (Excellent written and Spoken).
- Certification: FE – Civil (In Progress)

Education

Virginia Tech College of Engineering, Blacksburg, VA - Ph.D.

Graduated in **March 2025**

Key Research Areas:

- Stochastic Optimization of the trajectories of connected and autonomous vehicles in the vicinity of actuated signals considering surrounding traffic uncertain conditions.
- Vehicle queue length and traffic density estimation from Probe Vehicle Data using filtering algorithms such as Kalman Filters
- Development of a Game-theoretic based Traffic Signal controller.

Progress:

- Published two Journal papers at the IEEE Transactions on ITS (Q1).
- Published three conference papers at the IEEE Smart Mobility Conference in Niagara Falls, Canada, 2024.
- Published three conference papers at the Transportation Research Board (TRB) Annual Meetings 2023/2024/2025.

Cairo University, Faculty of Engineering, Cairo, Egypt - M.Sc.

Graduated in **December 2020**

Key Research Areas:

- Calibration of microscopic simulation models using machine learning models.
- Traffic survey analysis of a road network section in downtown Cairo, Egypt, and performed origin-Destination demand matrix estimation and calibration.

Achievements:

- Proposed Vissim calibration parameters to mimic observed driving behavior in the field.
- Published a research paper on research activities and findings.

Cairo University, Faculty of Engineering, Cairo, Egypt - B.Sc.

Graduated in **May 2016**

Specialization: Civil Engineering

Achievements:

- Graduated With Honors and appointed to join the faculty of engineering as a graduate teaching assistant.
- Received a funded scholarship to prepare for a master's degree.

Teaching Experience

Graduate Teaching Assistant

Virginia Tech, Department of Civil Engineering, Blacksburg, VA

Jan 2022 – Mar 2025

- Mentored the students of Traffic Engineering and Transportation Planning and Land Use courses during office hours through one-to-one communication.
- Grading Assignments for both courses.
- Demonstration and clarification of traffic analysis procedures to undergraduate and graduate students using HCM.
- Demonstrating Traffic simulation modeling to students.

Graduate Teaching Assistant

Cairo University, Faculty of Engineering, Cairo, Egypt

Jan 2017 – Aug 2021

- Demonstrated Assignments and solved problems related to the coursework for undergraduate students.
- Checked assignments, proctored tests, and provided grades according to university standards.

- Documented attendance and completed assignments to maintain full class and student.
- Prepared lessons according to the course outline to convey required material and deepen student understanding of the subject matter.
- Taught small groups of students focused on specific parts of the coursework.
- Mentored students through office hours and one-on-one communication.
- Assisted faculty with copying papers, collecting assignments, and preparing materials for upcoming classes.
- Helped in writing and submitting documents and reports associated with accreditation.

Research Experience

Graduate Research Assistant

Virginia Tech Transportation Institute (VTTI), Blacksburg, VA

Aug 2021 – Mar 2025

Conducting various analysis tasks, including:

- Extracting bike trajectories (speed profiles) from recorded campus videos to support the development of bike-following models.
- Processing large drone-based probe vehicle datasets to derive traffic demand and dynamics (e.g., vehicle speeds and accelerations), intended for advanced prediction and estimation algorithms such as Kalman filters and game-theoretic optimization.
- Optimizing connected and automated vehicle (CV/CAV) trajectories in stochastic environments near actuated traffic signals, accounting for traffic uncertainties.
- Estimating queue lengths and traffic density using probe vehicle data.
- Analyzing traffic performance metrics through advanced algorithms, including Kalman Filters.
- Optimizing traffic signals within a game-theoretic framework to enhance traffic flow and reduce delays.

Graduate Research Assistant

Cairo University, Faculty of Engineering, Cairo, Egypt

Jan 2017 – Aug 2021

- Worked on project teams to develop a strategy for e-mobility, and for the market potential assessment for the short to medium terms, advised by the (EBRD).
- Worked on calibrating Microscopic simulation parameters using machine learning on Vissim.
- Integrating Coding languages with PTV software through APIs.
- Using Genetic Algorithms to calibrate demand models for proposed railway networks.
- Performed sensitivity analysis of proposed scenarios for various railway projects and providing results for the economic feasibility team for evaluation.

Projects

Stochastic Optimization of Vehicle Trajectories, VT

- Use stochastic optimization methods and theories to implement a class project to optimize vehicle trajectories considering uncertainty in actuated traffic signals.

GIS Evaluation of Proposed Micromobility Stations Locations

- Worked with classmates to provide sufficient analytical evidence for implementing micro-mobility stations at vacant areas in Alexandria, VA.

Optimal Cost Policy for Inter-Campus Shared Mobility at Virginia Tech: A Transportation Problem Analysis

- Determined the optimal pricing policy for e-scooter mobility within the Virginia Tech campus area.
- Modeled multiple cost policies and calculated market share based on students' perceived cost of travel.
- Analysis highlighted the importance of aligning pricing with students' perceived value of time to maintain profitability.
- Used Gurobi for solving the optimization problem.

Traffic Analysis Highlight Projects in Egypt

- Abo Tartor Railway Extension Project.
- Gargoub Railway Extension Project.
- Portsaid Railway Extension Project.
- Taba Railway Extension Project.
- Ezz Steel Factories, Cairo, EG.
- Cairo-Aswan-Toshka Railway Extension Project.

- Economic Impacts of New Cairo Metro Projects (World Bank), Cairo, EG.
- ITS Deployment Plan for Cairo Ring Road and 26th July Corridor, Cairo, EG.
- E-Mobility Strategy and Market Study in Egypt, Cairo, EG.
- Al-Alamain Industrial City, EG.
- New Heliopolis City Masterplan, Cairo, EG.
- New Sphinx City Masterplan, Cairo, EG.
- Site visits and identification of accident hotspots in Giza Governorate, Giza, EG.
- Shebeen - Qanater Railway, Cairo, EG.
- Traffic Impact Assessment in Obour City, Cairo, EG.
- Traffic Impact Assessment on 10th of Ramadan City, Cairo, EG .
- Traffic Impact Assessment in Damietta City, Cairo, EG.
- Masterplan of Cairo Governorate Project, Cairo, EG.
- Historic Cairo Transport Network.
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Traffic Analysis Highlight Projects in KSA:

- Hassanaa Al Remal Neighbourhood Traffic Impact Study Riyadh.
- Al Nakheel Mall – New Extension, Riyadh.
- Khuzam Mall – Murcia, Riyadh.
- NEOM Hospital Traffic Impact Study, Neom.
- 6 - Drive Through Restaurants, Riyadh.
- Gamma 2.0 Traffic Impact Study, Riyadh.
- Azezia Fruits and Vegetable Market, Riyadh.
- Eteika Fruits and Vegetable Market, Riyadh.
- Mecca University Campus, Traffic Impact Assessment, Mecca.
- “Etlalet Mecca” compound, Traffic Impact Assessment, Mecca.
- A New School Traffic Impact Assessment, Riyadh.
- Riyadh Hotel Traffic Impact Assessment, Riyadh.
- Itqan Al Tarbeya School Traffic Impact Assessment, Riyadh.
- Medical Center Traffic Impact Assessment, Riyadh.
- A New School Traffic Impact Assessment, Riyadh.

Leadership and Volunteering

- STP – 2012: Volunteer in the Entrepreneurship Sessions.
- K-Vector 2013-2015: Moderator and head for the academic sessions that provides technical training and tutorials of engineering software for undergraduate civil engineering students.

Published Articles

Journal Articles

- A. K. Shafik**, and H. A. Rakha, "Optimization of Vehicle Trajectories Considering Uncertainty in Actuated Traffic Signal Timings," in *IEEE Transactions on Intelligent Transportation Systems*.
- A. K. Shafik**, S. Eteifa and H. A. Rakha, "Optimization of Vehicle Trajectories Considering Uncertainty in Actuated Traffic Signal Timings," in *IEEE Transactions on Intelligent Transportation Systems*, doi: 10.1109/TITS.2023.3253424
- A. K. Shafik**, and H. A. Rakha, "Real-Time Traffic State Estimation and Short-Term Prediction Using Probe Vehicle Data: A Kalman Filter Approach," in *Sensors*.
- A. K. Shafik**, Hassan, A., Saied, A. M., & Abdelmegeed, A. E. (2022). Development of Online Vissim Traffic Microscopic Calibration Framework Using Artificial Intelligence for Cairo CBD Area. *Ekistics and The New Habitat*, 81(1), 2-9. <https://doi.org/10.53910/26531313-E2021811505>

Conference Papers and Presentations

- A. K. Shafik**, Eteifa, S., & Rakha, H. (2023). Optimization of Vehicle Trajectories Considering Uncertainty in Actuated Traffic Signal Timings. Transportation Research Board (TRB) 102nd Annual Meeting
- A. K. Shafik** and H. A. Rakha, “Queue Estimation and Consideration in Vehicle Trajectory Optimization at Actuated Signalized Intersections,” in *IEEE Smart Mobility Conference 2024*.

- A. K. Shafik** and H. A. Rakha, “Kalman Filter-Based Real-Time Traffic State Estimation and Prediction Using Vehicle Probe Data,” in *IEEE Smart Mobility Conference 2024*.
- A. K. Shafik** and H. A. Rakha, “Enhancing and Evaluating a Decentralized Cycle-Free Game-Theoretic Adaptive Traffic Signal Controller on an Isolated Signalized Intersection,” in *IEEE Smart Mobility Conference 2024*.

Scholarships

The Last Mile Leadership Scholarship in the amount of \$6,000 for the 2024-2025 school year

References

Available Upon Request.